

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

December 2012

CL403 Design of Structures-I

Date: 06.12.2012, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.
5. IS 456:2000, IS 800:2007, SP 16, and steel table are permitted.

SECTION - I

Q - 1 What are dowels from footings? Why are they provided? [07]

A square footing $3 \text{ m} \times 3 \text{ m}$ and 600 mm thickness supports a $450 \text{ mm} \times 450 \text{ mm}$ column. Design the dowels between the column and the footings if the characteristic loads at the base of column are 780 kN live load and 1000 kN dead load. Take $f_y = 415 \text{ MPa}$, $f_{ck} = 20 \text{ MPa}$.

Q - 2 (a) Give any two differences between beam and slab. [04]

Determine the depth of one way slab required for flexure for the following data:

Effective span = 4 m, Live load = 2.5 kN/m^2 . Take $f_y = 415 \text{ MPa}$, $f_{ck} = 20 \text{ MPa}$.

(b) A simply supported rectangular beam of 4 m span carries a uniformly distributed [10]
characteristic load including self-weight of 20 kN/m . The beam section is $230 \text{ mm} \times 450 \text{ mm}$ overall. Design and show detailing of the beam. Check this beam for deflection and shear only. Take $f_y = 250 \text{ MPa}$, $f_{ck} = 20 \text{ MPa}$.

OR

Q - 2 (a) Check the simply supported one way slab for (i) Deflection, (ii) Shear, (iii) Cracking [04]
for the following data:

Effective span = 4 m, Effective depth = 160 mm, Main steel reinforcement = 448 mm^2 , Distribution steel = 192 mm^2 , Shear force = 25 kN. Take $f_y = 415 \text{ MPa}$, $f_{ck} = 20 \text{ MPa}$.

(b) Design and show detailing of a simply supported beam of effective span 8 m [10]
subjected to imposed loads of 35 kN/m . The beam section is $300 \text{ mm} \times 700 \text{ mm}$ overall. Check this beam for deflection only. Take $f_y = 415 \text{ MPa}$, $f_{ck} = 20 \text{ MPa}$, $f_{sc} = 350 \text{ MPa}$.

Q - 3 (a) Design a reinforced concrete slab of $6.3 \text{ m} \times 4.5 \text{ m}$ effective span supported on all the four sides. It has to carry a characteristics live load of 10 kN/m^2 in addition to its dead weight. Corners are not held down. Check this slab for control of deflection only. Also draw reinforcement details. Take $f_y = 415 \text{ MPa}$, $f_{ck} = 25 \text{ MPa}$. [05]

(b) Determine the reinforcement for a short column for the following data: [09]
 Column size = $400 \text{ mm} \times 600 \text{ mm}$, $P_u = 2000 \text{ kN}$, $M_{ux} = 160 \text{ kN-m}$, $M_{uy} = 120 \text{ kN-m}$.
 Take $f_y = 415 \text{ MPa}$, $f_{ck} = 25 \text{ MPa}$. Sketch the reinforcement details. Assume that moment due to minimum eccentricity is less than the value given above.

OR

Q - 3 (a) The part of a double storied office building is to be constructed in moderate exposure condition is shown in **Figure 1**. Design the slab S_5 for the following data: Live load = 4 kN/m^2 , Floor finish = 1.5 kN/m^2 , Floor to floor height = 3 m . Assume corners are held down. Take $f_y = 415 \text{ MPa}$, $f_{ck} = 25 \text{ MPa}$. [05]

(b) Design an axially loaded tied column $400 \text{ mm} \times 400 \text{ mm}$ pinned at both ends with an unsupported length of 3 m for carrying a factored load of 2300 kN . Take $f_y = 415 \text{ MPa}$, $f_{ck} = 20 \text{ MPa}$. Sketch the reinforcement details. Assume that minimum eccentricity is less than 0.05 times its lateral dimension. [09]

SECTION – II

Q - 4 Design a beam of 5 m effective span, carrying a uniform load of 20 kN/m , if the compression flange is laterally unsupported. Assume $f_y = 250 \text{ MPa}$. Give any one necessary check. [07]

Q - 5 (a) Design a tie member of roof truss subjected to working loads of 60 kN (Dead load) and 100 kN (Live load). Use double angle section connected back-to-back on either side of gusset 8 mm thick. Use bolted connection. Take $f_y = 250 \text{ MPa}$, $f_u = 410 \text{ MPa}$ for both member and bolt material. [07]

(b) Design a double angle section back-to-back on each side of gusset 10 mm thick for continuous principal rafter of a truss to carry factored load of 200 kN . Centre-to-centre length of member between centroids of connection is 2 m . [07]

OR

Q - 5 (a) Design a suitable angle section to carry tensile force of 200 kN . Use welded connections. [07]

(b) Design a column of I-section in a building, subjected to axial compressive force of [07]

800 kN. Height of column is 5 m with top end hinged and bottom end fixed. It is braced in order to prevent buckling about the weaker axis at a distance of 2 m from top end.

- Q - 6 (a) Give advantages and disadvantages of bolted connections. [03]
(b) Explain by means of sketches the failure of plates in a joint made with bearing bolts. [04]
(c) Differentiate 'Fillet weld' and 'Butt weld' by means of relevant sketches. [04]
(d) Explain the following terms with respect to 'fillet weld' by means of sketches: [03]
(i) Size of weld, (ii) Effective throat thickness, (iii) Effective length

OR

- Q - 6 (a) Give advantages and disadvantages of welded connections. [03]
(b) Differentiate 'Plug weld' and 'slot weld' by means of relevant sketches. [04]
(c) Differentiate 'Lap joint' and 'Butt joint' with reference to bolted connections by means of relevant sketches. [04]
(d) Show by means of sketch, what you understand by the following terms with reference to bolted connections: [03]
(i) Gauge distance, (ii) Pitch of the bolts, (iii) Edge distance



Figure 1
